Input power connector
J1
ADAPTER 4.75–21V
F1
0451005.MRL 5A
Battery connector
J3
BATTERY 2S/4S
Output power connector
J2
VSYS OUTPUT
IC1
BQ25798RQMR
Q1
AON7528
Q2
AON7528
D1
12V
R1
270R
boost side
buck side
L1
2.2uH
J4
MCU / BIAS INTERFACE
NTC temp sensor
J6
Grounds
NORMAL 3V3 / SERVICE USB 3V3
U3
TPS62933DRLR NORMAL 3V3
U1
ESP32-C3-MINI-1-N4X
U4
AP7361C-33E-13 USB 3V3 LDO
POWER MUX - MAIN PRIORITY
U5
TPS2116DRL NORMAL PRIORITY
I2C / INT DOMAIN ISOLATION
U6
TCA9517DGKR REV.0
Q8
PMV16UN MAIN_VALID
Q9
PMV16UN TCA_EN
CHARGING: MAIN PRESENT AND FIRMWARE REQUEST
Q4
BC847 INT DOMAIN ISOLATOR
Q6
BC847 MAIN_VALID SINK
Q7
BC847 MCU CHG ARM SINK
Q5
BC857 MAIN-GATED CHG SOURCE
USB-C PROGRAM / NATIVE USB
U2
TPD2EUSB30A (PROVISIONAL)
SW1
BOOT / DOWNLOAD
SW2
RESET
U7
TPS62933DRLR
U8
LM74700QDBVRQ1
Q10
J8
5V / GND

F1 prototype: Littelfuse 0451005.MRL, 5 A, 125 Vdc; qualify temperature, cable, hot-plug, polarity-FET SOA and coordination.

VIN damping R12 2 ohm / C26 2.2uF 100V: selected prototype parts; VAC1 hot-plug qualification OPEN.

The REGN LDO supplies internal bias circuits and the MOSFETs gate drivers.
The pull-up rails of ILIM_HIZ, TS, and STAT can be connected to REGN.
The REGN is not recommended to source any other external circuit!

PGND = system return; GND = quiet TS / PROG / NTC; NT1 local tie; no PCB moat.

VIN hot-plug RC: TI Fig. 8-1 (>15V)
Provisional R12 / C26; validate on prototype

PMID bulk C20/C22/C24/C25: DNP in initial VSYS profiles; backup bulk is deferred.

Q5/Q6/Q7 default off; Q3 sinks CE only on request. Q6 is not a precision MAIN voltage monitor.

CE pull-up via REGN/MAIN diode OR. BQ25792 2S charger-only reset/default 2A risk OPEN.

SERVICE USB / ESP32-C3 RECOVERY – prototype, not fabrication approval

U3 SYS –> MAIN_3V3 buck; U4 USB_VBUS –> USB3V3 LDO; U5 normal-priority mux –> MCU3V3.

USB-only ROM flashing: BOOT held low during RESET; no battery/adapter required. Wi-Fi off in service firmware.

Complete power interruption: EN 10k/1u RC and GPIO straps restart MCU. Transition resets acceptable; hardware tests pending.

U1 project footprint corrected to 61 lands / 53 numbers; verify RF keepout, stencil and placement in PCB phase.

POGO SERVICE: TP5 TX, TP6 RX, TP7 GND, TP8 MCU3V3 SENSE, TP9 RESET, TP10 BOOT.

PCB edge, proposed 2.54mm pitch, TOP view pin1 triangle; fixture high-Z until target MCU3V3 valid.

TP11/12 MCU I2C; TP13 MAIN3V3. TP14 CE_N: passive probe only, separate charger-domain pad.

POWER MEASUREMENTS: TP15 VIN1, TP16 BATTERY, existing TP4 SYS; separate from MCU pogo group.

TP17 F5 OUTPUT: measurement only. No testpad is an authorized power input.

F5 OPTIONAL 5V / 2A: populate ALL U7 U8 Q10 L3 R60–R63 C60–C69 J8; F0 omit all.

SYS 6–18V –> 5.04V buck –> ideal diode –> J8 pin1 +5V, pin2 PGND. No MCU/USB control.

Sheet: /
File: ups12v_v4.kicad_sch

Title: BQ25798 UPS module

Size: A1
Date:
Rev: 2.1
KiCad E.D.A. 10.0.7
Id: 1/1